WINNER RANKINGS

1	Horseshoe False Discovery Rate Control via Frequentist Thresholding Im	65%
2	T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co-	63%
3	Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Ca	62%
4	**GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyon** ◀	**62%**
5	Conditional Independence Masking in JEPA Targets Suppresses Co-express	60%

GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyond Co-expression

Status: Winner · **Domain:** Gene Regulatory Network Inference / Self-Supervised Representation Learning · **Generation:** 0 · **Score:** 62%

Abstract

A fundamental failure mode in two-stage gene regulatory network (GRN) inference is that the Stage 1 encoder learns co-expression patterns rather than causal regulatory relationships, causing the weight matrix W_init to carry insufficient regulatory signal even when MSE loss converges. This occurs because standard JEPA objectives optimize prediction in latent space without any biological grounding, allowing the encoder to exploit co-expression shortcuts. Current literature on self-supervised learning shows that auxiliary grounding mechanisms—specifically Gaussian Mixture Model soft posteriors as anchor targets—prevent representation collapse and direct encoders toward semantically meaningful structure [8] . Meanwhile, GRN inference literature establishes that co-expression and causation are empirically distinguishable via conditional independence patterns [38, 59]. We hypothesize that replacing standard JEPA prediction targets with GMM soft posteriors derived from known TF binding motif co-occurrence clusters will force W_init to encode regulatory cluster membership rather than raw co-expression, measurably increasing W_init AUROC on held-out validation edges above the 0.58 threshold. The methodology involves fitting a k=20–40 GMM on TF-gene expression profiles augmented with motif co-occurrence distances, then training Stage 1 with a dual MSE + KL-divergence loss against frozen GMM posteriors. Expected outcome: W_init AUROC ≥ 0.62 on net3, enabling genuine two-stage cooperation and net3 AUPR > 0.099.

1. INTRODUCTION & SIGNIFICANCE

Paper [8] (GMM-Anchored JEPA) demonstrates that fitting a GMM once on input features and using its frozen soft posteriors as auxiliary targets prevents representation collapse in JEPA and grounds representations in semantically meaningful clusters, outperforming unanchored JEPA on downstream tasks. Paper [38] establishes that GRN inference is confounded by latent variables and co-expression, and that methods lacking explicit causal grounding systematically conflate correlation with regulation. The gap is that no existing JEPA variant for tabular/genomic data uses domain-specific motif co-occurrence structure as the GMM input, leaving the encoder free to learn co-expression shortcuts.

2. HYPOTHESIS STATEMENT

> *Replacing standard JEPA latent prediction targets with frozen GMM soft posteriors derived from motif-co-occurrence-augmented TF-gene profiles will increase W_init AUROC on held-out regulatory edges from below 0.58 to above 0.62 on net3 (E.coli), measurably distinguishing regulatory from co-expression signal.*

3. BACKGROUND & LITERATURE REVIEW

- **Soft Clustering Anchors for Self-Supervised Speech Representation Learning in Joint Embedding Prediction Architectures** — Ioannides et al. (2026)
- **Gene Regulatory Network Inference in the Presence of Selection Bias and Latent Confounders** — Luo et al. (2025)
- **Gene Regulatory Network Inference in the Presence of Dropouts: a Causal View** — Dai et al. (2024)
- **T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data** — Thimonier et al. (2024)
- **Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architecture -- Bridging Predictive and Generative Self-Supervised Learning** — Gögl et al. (2026)
- **Machine Learning Methods for Gene Regulatory Network Inference** — Hegde et al. (2025)
- **Learning Active Subspaces and Discovering Important Features with Gaussian Radial Basis Functions Neural Networks** — D'Agostino et al. (2023)
- **ST-GCP: a graph convolutional network model with contrastive consistency and permutation for spatial transcriptomics.** — Meng et al. (2025)
- *+ 18 additional papers — see Section 8.*

4. METHODOLOGY

Fit a GMM on a combined feature matrix of TF-gene expression profiles and pairwise TF binding motif co-occurrence distances for each DREAM5 network. Use frozen GMM soft posteriors as auxiliary targets during Stage 1 JEPA training alongside the standard MSE prediction loss. Evaluate W_init AUROC on held-out regulatory edges before passing to Stage 2 horseshoe regression. Compare net3 AUPR to the GENIE3 baseline.

Experimental Phases

1. Construct a combined feature matrix for each DREAM5 network by concatenating normalized TF-gene expression profiles with pairwise TF binding motif co-occurrence distance vectors (using JASPAR or RegulonDB motif databases for net3/net2).
2. Fit a GMM with $k \in \{20, 30, 40\}$ components on this combined matrix using EM; freeze the GMM parameters and precompute soft posterior probabilities $q(z|x)$ for every TF-gene pair.
3. Modify the Stage 1 JEPA training loop to add a KL-divergence auxiliary loss against frozen GMM posteriors, weighted by λ_gmm $\in \{0.1, 0.5, 1.0\}$; train for 150–300 epochs (feasible at <3s/10 epochs).
4. Extract W_init after Stage 1 and compute AUROC against held-out positive/negative regulatory edge labels from the DREAM5 gold standard for each network; target AUROC > 0.58 on net3.
5. Pass W_init to Stage 2 horseshoe regression and train for 200 epochs; record net3 AUPR and compare to GENIE3 baseline (0.099) and baseline W_init-only AUPR.
6. Ablate: run Stage 1 with GMM anchoring but without motif co-occurrence (expression-only GMM) to isolate the motif contribution to W_init AUROC.

5. EXPECTED OUTCOMES

Predicted Impact: W_init AUROC improvement from <0.58 to ≥0.62 on net3; net3 AUPR improvement from ~0.090 (current estimate) to >0.099 (beating GENIE3); potential 10–20% AUPR gain on net2 (small-n regime where regulatory grounding matters most).

Success Criteria: W_init AUROC > 0.58 on all four DREAM5 networks and > 0.62 on net3; Stage 2 net3 AUPR > 0.099; net2 AUPR maintained above 0.0164; total Stage 1 training time remains < 3s per 10 epochs on Apple Silicon MPS.

Null Result Interpretation: If this hypothesis is false: Instead of frozen GMMs, use a supervised contrastive loss directly on known-regulatory vs. known-non-regulatory TF-gene pairs from gold-standard subsets, anchored by motif affinity as a continuous auxiliary weight, to create a target that is explicitly aligned with causal regulation rather than clustering structure.

6. RISK ANALYSIS & MITIGATION

Fatal Assumptions

- ☠ GMM clusters derived from motif-co-occurrence-augmented TF-gene profiles genuinely separate regulatory from co-expression signal — if false: the auxiliary target is just a noisy re-labeling of co-expression clusters, and AUROC improvement is expected to be zero or negative, making the entire Stage 1 modification useless.
- ☠ E. coli motif database coverage is sufficient to provide informative pairwise motif co-occurrence distances for a substantial fraction of net3 TF-gene pairs — if false: the GMM is fitted on sparse/zero motif columns, the clustering is driven purely by expression, and the hypothesis collapses to standard co-expression-based JEPA.

Weakest Assumption: That GMM soft posteriors on motif-augmented profiles separate causal regulatory pairs from co-expressed pairs — if the GMM clusters are dominated by expression covariance, the auxiliary target provides no causal disambiguation.

Key Risks

- GMM soft posteriors derived from expression + motif co-occurrence may still cluster co-regulated genes rather than TF→gene causal pairs, meaning the auxiliary target encodes co-expression topology and the AUROC gain over baseline may be negligible or even negative.
- Motif co-occurrence distances are a proxy for physical binding, not functional regulation; for E. coli (net3) the motif database coverage is incomplete, so many true regulatory pairs will have zero motif signal, causing the GMM to assign them to the same uninformative cluster.
- Jointly fitting a GMM on expression and motif features requires careful feature scaling; if expression variance dominates, motif distances are effectively ignored, degenerating to the original co-expression-only JEPA target.
- Frozen GMM posteriors from the full training set leak information about held-out validation edges if any training samples overlap with the validation split, inflating AUROC estimates artificially.

Minimal Validation Experiment

Before full JEPA training: fit the GMM on E. coli (net3) expression+motif features, then compute AUROC of the soft posterior assignment (highest-posterior cluster identity) as a standalone edge predictor on known regulatory edges vs. randomly sampled non-edges. Takes ~1 day. If AUROC < 0.58, the GMM target carries no regulatory signal and the hypothesis is dead; if AUROC ≥ 0.62, proceed to full JEPA training.

Competing Mechanisms

- ⇄ Alternative 1 (Expression-Only Clustering): The improvement in W_init AUROC comes purely from clustering structure in expression space rather than motif co-occurrence — if true, a GMM fit on expression data alone would achieve identical AUROC to the motif-augmented GMM, and the motif ablation would show no difference from the full model.
- ⇄ Alternative 2 (Stage 2 Dominance): W_init AUROC is irrelevant to final AUPR because horseshoe regression in Stage 2 effectively re-learns regulatory weights from scratch regardless of initialization — if true, randomizing W_init would produce equivalent or identical Stage 2 AUPR, making any Stage 1 improvement statistically indistinguishable from baseline.

Discriminating Experiment

Run three Stage 2 training runs with identical horseshoe hyperparameters but three different W_init sources: (1) GMM-Anchored JEPA with motif co-occurrence, (2) GMM-Anchored JEPA with expression-only GMM, (3) random W_init (Glorot uniform). Measure net3 AUPR for all three. If THIS hypothesis is correct: motif-augmented GMM > expression-only GMM > random, with motif-augmented achieving AUPR > 0.099. If Alternative 1 is correct: motif-augmented ≈ expression-only > random. If Alternative 2 is correct: all three converge to the same AUPR within noise.

7. LIMITATIONS & ASSUMPTIONS

Key Assumptions (most → least confident)

1. Assumption [85%]: Stage 1 runs fast enough (<3s/10 epochs) even with the added GMM KL loss computation — Risk: precomputing soft posteriors for all TF-gene pairs may add memory overhead on MPS, though the GMM is frozen so no gradient flows through it.
2. Assumption [75%]: TF binding motif co-occurrence data is available for net3 (E.coli) and net2 (S.aureus) at sufficient coverage — Risk: S.aureus motif databases are less complete than E.coli, potentially leaving net2 with sparse co-occurrence features.
3. Assumption [65%]: GMM soft posteriors derived from motif co-occurrence capture regulatory cluster membership that is informative for edge prediction beyond what expression alone provides — Risk: motif co-occurrence may be too coarse or noisy to add signal beyond expression-based clustering.
4. Assumption [50%]: W_init AUROC > 0.58 is causally necessary (not merely correlated) for Stage 2 AUPR > 0.099 — Risk: Stage 2 horseshoe may be robust to initialization quality, making W_init improvement irrelevant to final AUPR (Alternative 2 being true).

Major Assumptions — require revision if false

- ⚠ Joint feature scaling of expression and motif distances preserves a meaningful motif contribution to cluster assignments — if false: expression variance dominates, motif signal is negligible, and the GMM component proportions need to be redesigned (e.g., separate GMMs per modality with late fusion).
- ⚠ Frozen GMM posteriors remain stable predictors of regulatory membership as Stage 1 encoder weights update — if false: iterative re-fitting of the GMM would be needed, significantly increasing computational cost.
- ⚠ The JEPA encoder's weight matrix W_init meaningfully extracts the regulatory information embedded in the GMM targets rather than fitting the cluster structure independently of regulation — if false: the AUROC improvement on W_init may not translate to Stage 2 AUPR gains.

8. EVIDENCE SUMMARY

DIMENSION	SCORE	WEIGHT
EVIDENCE GROUNDING	62%	35%
Feasibility	55%	30%
Impact Potential	65%	25%
Novelty	72%	10%
Composite	**62%**	weighted

Why this won: The core problem—Stage 1 learning co-expression not causation—maps directly to the GMM-Anchored JEPA mechanism from paper [8] , which was designed precisely to prevent representation collapse and ground encoders in semantically meaningful clusters. The novelty is applying domain-specific motif co-occurrence as the GMM input, making the anchor targets biologically meaningful for GRN inference. This is feasible on Apple Silicon (Stage 1 is fast), directly addresses the AUROC > 0.58 success criterion, and produces a testable, falsifiable prediction (AUROC change) that can be measured before the expensive Stage 2 run.

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[2] Luo, G., Dai, H., Li, L., Gao, C., & Sun, B. (2025). *Gene Regulatory Network Inference in the Presence of Selection Bias and Latent Confounders*.

[3] Dai, H., Ng, I., Luo, G., Spirtes, P., & Stojanov, P. (2024). *Gene Regulatory Network Inference in the Presence of Dropouts: a Causal View*.

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